

Modül 31

Angular Material

Material Design 3 (M3) — Resmi Component Library

Modül: 31 / 30

Ders Sayısı: 8 ders

Seviye: Orta

Versiyon: Angular v21

Hazırlayan: Claude • Türkçe Angular v21 Eğitimi

☐ Modül 31 (BONUS) - Angular Material

Bu modül Material Design 3 (M3) ile Angular Material v20+'ı kapsıyor. Angular ekibinin kendi geliştirdiği resmi UI library — framework ile birlikte evrim geçirir. Material'ın temel kurulumundan production kullanımına kadar tüm adımları işleyeceğiz.

Bu modülün sonunda yapabileceklerin:

- ✓ Angular Material v20+ kurulum
- ✓ Material Design 3 (M3) theme sistemi
- ✓ Temel component kullanımı (Button, Form Field, Card, vb.)
- ✓ Form'larda Material entegrasyonu
- ✓ Dialog, Snackbar, BottomSheet servisleri
- ✓ Navigation (Toolbar, Sidenav, Tabs)
- ✓ Theming ve custom palette
- ✓ Angular CDK temelleri

Material Design 3 (M3) Nedir?

Google'ın 2021 sonrası tasarım dili. Eski MDC'den farkı.

Material Design Evrimi

Versiyon	Yıl	Karakter
M1	2014	Düz tasarım, güçlü renkler
M2	2018	Esnek, brand-friendly
M3 (Material You)	2021+	Dynamic colors, adaptif

M3'ün Yeni Konseptleri

- ✓ Dynamic Color: Renk paletleri image'dan veya seed renkten türetilir
- ✓ Color Roles: primary, secondary, tertiary, surface — rol-based
- ✓ Tonal Variants: Her renk için 13 ton (0-100)
- ✓ Adaptive Layouts: Mobile/tablet/desktop için farklı davranış
- ✓ Larger Touch Targets: Erişilebilirlik öncelikli

Angular Material v20'de M3

Angular Material v17+ ile M3 desteği eklendi. v20'de M3 default oldu. Eski M2 hâlâ destekleniyor ama yeni projeler M3 ile başlamalı.

Önemli Not

Eğer M2'li eski bir tutorial okuyorsan dikkat — API'lar değişti. Bu modül M3 odaklı.

Kurulum

Sıfırdan Angular projesine Material eklemek.

⚙️ Yeni Proje + Material

```
# Yeni Angular projesi
ng new my-app --style=scss --routing
cd my-app

# Material kur
ng add @angular/material

# Soruları yanıtla:
# ? Choose a prebuilt theme name, or 'custom':
#   - Azure/Blue (M3)           ← Modern, önerilen
#   - Rose/Red (M3)
#   - Magenta/Violet (M3)
#   - Cyan/Orange (M3)
#   - Custom theme
# ? Set up global Angular Material typography styles? Yes
# ? Include the Angular animations module? Include and enable animations
```

📦 Otomatik Yapılan Değişiklikler

```

package.json
+ @angular/material      ← Component library
+ @angular/cdk           ← Component Dev Kit (zorunlu peer)
+ @angular/animations    ← Animasyonlar (zorunlu)

src/styles.scss
+ @use '@angular/material' as mat;
+ html {
+   @include mat.theme((
+     color: (
+       theme-type: light,
+       primary: mat.$azure-palette,
+       tertiary: mat.$blue-palette,
+     ),
+     typography: Roboto,
+     density: 0,
+   ));
+ }

src/index.html
+ <link href="...Roboto..." rel="stylesheet"> ← Font
+ <link href="...Material+Symbols..." rel="stylesheet"> ← Icons
+ <body class="mat-typography">

```

□ Test Etme

```

// app.component.ts
import { Component } from '@angular/core';
import { MatButtonModule } from '@angular/material/button';
import { MatIcon } from '@angular/material/icon';

@Component({
  selector: 'app-root',
  imports: [MatButtonModule, MatIcon],
  template: `
    <button matButton="filled">Hello Material!</button>
    <button matButton="outlined">
      <mat-icon>favorite</mat-icon>
      Like
    </button>
  `,
})
export class AppComponent {}

```

⚠ v20 Button API Değişikliği

Eski: mat-button, mat-raised-button. Yeni: matButton="filled",
 matButton="outlined". Eski tutorial'larda eski syntax görebilirsin.

Temel Komponentler

Button, Card, Icon, Chip — en sık kullanılanlar.

□ Button

```
import { MatButtonModule, MatIconButton, MatFabButton } from '@angular/material/button';

@Component({
  imports: [MatButtonModule, MatIconButton, MatFabButton],
  template: `
    <!-- Text button (M3 default) -->
    <button matButton>Text</button>

    <!-- Filled (eski: raised) -->
    <button matButton="filled">Filled</button>

    <!-- Tonal -->
    <button matButton="tonal">Tonal</button>

    <!-- Outlined (eski: stroked) -->
    <button matButton="outlined">Outlined</button>

    <!-- Elevated -->
    <button matButton="elevated">Elevated</button>

    <!-- Icon button -->
    <button matIconButton aria-label="Favori">
      <mat-icon>favorite</mat-icon>
    </button>

    <!-- FAB (Floating Action Button) -->
    <button matFab>
      <mat-icon>add</mat-icon>
    </button>

    <!-- Extended FAB -->
    <button matFab extended>
      <mat-icon>edit</mat-icon>
      Düzenle
    </button>

    <!-- Color -->
    <button matButton="filled" color="primary">Primary</button>
    <button matButton="filled" color="warn">Sil</button>
  `,
})
```

★ Card

```
import { MatCardModule } from '@angular/material/card';

<mat-card>
  <mat-card-header>
    <div mat-card-avatar class="my-avatar"></div>
    <mat-card-title>Shiba Inu</mat-card-title>
    <mat-card-subtitle>Köpek Cinsi</mat-card-subtitle>
  </mat-card-header>

  

  <mat-card-content>
    <p>Shiba Inu Japonya'nın küçük cinslerinden biri...</p>
  </mat-card-content>

  <mat-card-actions>
    <button matButton>LIKE</button>
    <button matButton>SHARE</button>
  </mat-card-actions>
</mat-card>
```

□ Icon

```
import { MatIcon } from '@angular/material/icon';

<!-- Material Symbols (modern, default) -->
<mat-icon>home</mat-icon>
<mat-icon>favorite</mat-icon>
<mat-icon>account_circle</mat-icon>
<mat-icon>delete</mat-icon>
<mat-icon>settings</mat-icon>

<!-- SVG icon (custom) -->
<mat-icon svgIcon="custom-logo"></mat-icon>

<!-- Renkli -->
<mat-icon color="primary">star</mat-icon>
<mat-icon color="warn">error</mat-icon>

<!-- Filled/outlined variant -->
<mat-icon fontIcon="favorite"></mat-icon>
```

□ Chip

```
import { MatChipsModule } from '@angular/material/chips';

<mat-chip-set>
  <mat-chip>Angular</mat-chip>
  <mat-chip>TypeScript</mat-chip>
  <mat-chip>Material</mat-chip>
</mat-chip-set>

<!-- Removable -->
<mat-chip-listbox>
  @for (tag of tags(); track tag) {
    <mat-chip-option (removed)="remove(tag)">
      {{ tag }}
      <button matChipRemove>
        <mat-icon>cancel</mat-icon>
      </button>
    </mat-chip-option>
  }
</mat-chip-listbox>
```


Form Components

Form Field, Input, Select, DatePicker — kullanıcı girdileri.

□ Form Field + Input

```
import { MatFormField, MatLabel, MatHint, MatError } from
 '@angular/material/form-field';
import { MatInput } from '@angular/material/input';

<mat-form-field appearance="outline">
  <mat-label>Email</mat-label>
  <input matInput type="email" [(value)]="email" required>
  <mat-hint>Geçerli bir email girin</mat-hint>
  @if (emailControl.hasError('required')) {
    <mat-error>Email zorunlu</mat-error>
  }
</mat-form-field>

<!-- Prefix / Suffix -->
<mat-form-field appearance="outline">
  <mat-label>Şifre</mat-label>
  <input matInput [type]="hidePassword() ? 'password' : 'text'">
  <mat-icon matPrefix>lock</mat-icon>
  <button matIconButton matSuffix (click)="togglePassword()">
    <mat-icon>{{ hidePassword() ? 'visibility' : 'visibility_off' }}</mat-icon>
  </button>
</mat-form-field>
```

□ Select

```

import { MatSelectModule } from '@angular/material/select';

<mat-form-field appearance="outline">
  <mat-label>Şehir</mat-label>
  <mat-select [(value)]="selectedCity">
    @for (city of cities; track city.value) {
      <mat-option [value]="city.value">{{ city.label }}</mat-option>
    }
  </mat-select>
</mat-form-field>

<!-- Multi-select -->
<mat-form-field appearance="outline">
  <mat-label>Beceriler</mat-label>
  <mat-select [(value)]="skills" multiple>
    @for (skill of allSkills; track skill) {
      <mat-option [value]="skill">{{ skill }}</mat-option>
    }
  </mat-select>
</mat-form-field>

<!-- Gruplu -->
<mat-select>
  @for (group of groups; track group.name) {
    <mat-optgroup [label]="group.name">
      @for (item of group.items; track item.value) {
        <mat-option [value]="item.value">{{ item.label }}</mat-option>
      }
    </mat-optgroup>
  }
</mat-select>

```

□ DatePicker

```

import { MatDatepickerModule } from '@angular/material/datepicker';
import { provideNativeDateAdapter } from '@angular/material/core';

// app.config.ts veya component
providers: [provideNativeDateAdapter()]

<mat-form-field appearance="outline">
  <mat-label>Doğum tarihi</mat-label>
  <input matInput [matDatepicker]="picker" [(value)]="birthDate">
  <mat-datepicker-toggle matIconSuffix [for]="picker"></mat-datepicker-toggle>
  <mat-datepicker #picker></mat-datepicker>
</mat-form-field>

<!-- Date range -->
<mat-form-field appearance="outline">
  <mat-label>Tarih aralığı</mat-label>
  <mat-date-range-input [rangePicker]="rangePicker">
    <input matStartDate placeholder="Başlangıç">
    <input matEndDate placeholder="Bitiş">
  </mat-date-range-input>
  <mat-datepicker-toggle matIconSuffix [for]="rangePicker"></mat-datepicker-toggle>
  <mat-date-range-picker #rangePicker></mat-date-range-picker>
</mat-form-field>

```

☑ Checkboxes & Radio & Slider

```

import { MatCheckbox } from '@angular/material/checkbox';
import { MatRadioModule } from '@angular/material/radio';
import { MatSlider, MatSliderThumb } from '@angular/material/slider';
import { MatSlideToggle } from '@angular/material/slide-toggle';

<!-- Checkbox -->
<mat-checkbox [(checked)]="accept">Kabul ediyorum</mat-checkbox>

<!-- Slide Toggle -->
<mat-slide-toggle [(checked)]="darkMode">Karanlık mod</mat-slide-toggle>

<!-- Radio Group -->
<mat-radio-group [(value)]="selectedSize">
  <mat-radio-button value="S">Küçük</mat-radio-button>
  <mat-radio-button value="M">Orta</mat-radio-button>
  <mat-radio-button value="L">Büyük</mat-radio-button>
</mat-radio-group>

<!-- Slider -->
<mat-slider min="0" max="100" step="5">
  <input matSliderThumb [(value)]="volume">
</mat-slider>

```

Overlay Services

Dialog, Snackbar, BottomSheet — runtime UI elementleri.

□ Dialog

```

import { MatDialog, MatDialogRef, MAT_DIALOG_DATA } from '@angular/material/dialog';
import { inject } from '@angular/core';

// Dialog component
@Component({
  selector: 'app-confirm-dialog',
  imports: [MatDialogModule, MatButtonModule],
  template: `
    <h2 matDialogTitle>{{ data.title }}</h2>
    <mat-dialog-content>
      <p>{{ data.message }}</p>
    </mat-dialog-content>
    <mat-dialog-actions align="end">
      <button matButton matDialogClose>İptal</button>
      <button matButton="filled" color="warn" [matDialogClose]="true">
        Onayla
      </button>
    </mat-dialog-actions>
  `,
})
export class ConfirmDialog {
  data = inject<{ title: string; message: string }>(MAT_DIALOG_DATA);
}

// Çağırın component
@Component({...})
export class UserListComponent {
  private dialog = inject(MatDialog);

  async deleteUser(user: User) {
    const result = await firstValueFrom(
      this.dialog.open(ConfirmDialog, {
        width: '400px',
        data: {
          title: 'Kullanıcıyı sil',
          message: `${user.name} kullanıcısı silinecek. Emin misin?`
        }
      }).afterClosed()
    );

    if (result === true) {
      // Sil
    }
  }
}

```

□ SnackBar (Toast)

```
import { MatSnackBar } from '@angular/material/snack-bar';

@Component({...})
export class MyComponent {
  private snackBar = inject(MatSnackBar);

  showSuccess() {
    this.snackBar.open('Başarıyla kaydedildi!', 'Tamam', {
      duration: 3000,
      horizontalPosition: 'right',
      verticalPosition: 'top',
      panelClass: ['success-snackbar'],
    });
  }

  // Action ile
  showUndo() {
    const ref = this.snackBar.open('Silindi', 'GERİ AL', {
      duration: 5000,
    });

    ref.onAction().subscribe(() => {
      // Geri al
    });
  }
}
```

□ BottomSheet

```
import { MatBottomSheet, MatBottomSheetRef } from '@angular/material/bottom-sheet';

@Component({
  template: `
    <mat-nav-list>
      <a mat-list-item (click)="select('share')">
        <mat-icon matListItemIcon>share</mat-icon>
        <span matListItemTitle>Paylaş</span>
      </a>
      <a mat-list-item (click)="select('copy')">
        <mat-icon matListItemIcon>content_copy</mat-icon>
        <span matListItemTitle>Kopyala</span>
      </a>
    </mat-nav-list>
  `,
})
export class ShareSheet {
  private ref = inject<MatBottomSheetRef<ShareSheet>>(MatBottomSheetRef);

  select(option: string) {
    this.ref.dismiss(option);
  }
}

// Açma
const sheetRef = this.bottomSheet.open(ShareSheet);
sheetRef.afterDismissed().subscribe(result => {
  console.log('Selected:', result);
});
```

Navigation Components

Toolbar, Sidenav, Tabs, Menu, Stepper.

☐ Toolbar

```
import { MatToolbar } from '@angular/material/toolbar';

<mat-toolbar color="primary">
  <button matIconButton (click)="toggleSidenav()">
    <mat-icon>menu</mat-icon>
  </button>

  <span>My App</span>

  <span class="spacer" style="flex: 1 1 auto;"></span>

  <button matIconButton>
    <mat-icon>notifications</mat-icon>
  </button>

  <button matIconButton [matMenuTriggerFor]="userMenu">
    <mat-icon>account_circle</mat-icon>
  </button>

  <mat-menu #userMenu="matMenu">
    <button matMenuItem>Profil</button>
    <button matMenuItem>Ayarlar</button>
    <button matMenuItem (click)="logout()">Çıkış</button>
  </mat-menu>
</mat-toolbar>
```

☐ Sidenav


```

import { MatSidenavModule } from '@angular/material/sidenav';
import { MatListModule } from '@angular/material/list';

<mat-sidenav-container class="container">
  <mat-sidenav #drawer mode="side" opened>
    <mat-nav-list>
      <a mat-list-item routerLink="/home" routerLinkActive="active">
        <mat-icon matListItemIcon>home</mat-icon>
        <span matListItemTitle>Ana Sayfa</span>
      </a>
      <a mat-list-item routerLink="/users" routerLinkActive="active">
        <mat-icon matListItemIcon>people</mat-icon>
        <span matListItemTitle>Kullanıcılar</span>
      </a>
    </mat-nav-list>
  </mat-sidenav>

  <mat-sidenav-content>
    <router-outlet />
  </mat-sidenav-content>
</mat-sidenav-container>

```

☐ Tabs

```

import { MatTabsModule } from '@angular/material/tabs';

<mat-tab-group>
  <mat-tab label="Profil">
    <app-profile />
  </mat-tab>
  <mat-tab label="Ayarlar">
    <app-settings />
  </mat-tab>
  <mat-tab label="Bildirimler">
    <app-notifications />
  </mat-tab>
</mat-tab-group>

<!-- Lazy load tab content -->
<mat-tab-group>
  <mat-tab label="İlk tab">
    <ng-template matTabContent>
      <!-- Sadece tab aktif olunca render -->
      <app-heavy-content />
    </ng-template>
  </mat-tab>
</mat-tab-group>

```

☐ Stepper

```
import { MatStepperModule } from '@angular/material/stepper';

<mat-stepper linear #stepper>
  <mat-step [stepControl]="form1" label="Hesap bilgileri">
    <form [formGroup]="form1">
      <!-- Form fields -->
    </form>
    <div>
      <button matButton matStepperNext>İleri</button>
    </div>
  </mat-step>

  <mat-step [stepControl]="form2" label="Adres">
    <form [formGroup]="form2">
      <!-- Form fields -->
    </form>
    <div>
      <button matButton matStepperPrevious>Geri</button>
      <button matButton="filled" matStepperNext>İleri</button>
    </div>
  </mat-step>

  <mat-step label="Onay">
    <p>Tüm bilgiler doğru mu?</p>
    <button matButton matStepperPrevious>Geri</button>
    <button matButton="filled" color="primary" (click)="submit()">Tamamla</button>
  </mat-step>
</mat-stepper>
```

M3 Theming & Custom Palette

Markana özel renkler ve dark mode.

□ Built-in Palette'lar

```
// styles.scss
@use '@angular/material' as mat;

html {
  @include mat.theme((
    color: (
      theme-type: light,           // light | dark
      primary: mat.$azure-palette, // ana renk paleti
      tertiary: mat.$blue-palette, // tertiary
    ),
    typography: Roboto,           // veya custom config
    density: 0,                  // -3 (compact) to 0 (default)
  ));
}

// Built-in palette'lar:
// $azure-palette, $blue-palette, $cyan-palette,
// $green-palette, $magenta-palette, $orange-palette,
// $red-palette, $rose-palette, $spring-green-palette,
// $violet-palette, $yellow-palette
```

□ Dark Mode

```
// Yöntem 1: Sistem tercihine göre otomatik
html {
  @include mat.theme((
    color: (
      theme-type: light,
      primary: mat.$azure-palette,
    ),
  ));
}

@media (prefers-color-scheme: dark) {
  html {
    @include mat.theme((
      color: (
        theme-type: dark,
        primary: mat.$azure-palette,
      ),
    ));
  }
}

// Yöntem 2: Manuel toggle (class ile)
html {
  @include mat.theme((color: (theme-type: light, primary: mat.$azure-palette)));
}

html.dark-mode {
  @include mat.theme((color: (theme-type: dark, primary: mat.$azure-palette)));
}
```

□ Component'te Toggle

```

@Injectables({ providedIn: 'root' })
export class ThemeService {
  isDark = signal(false);

  constructor() {
    // İlk yüklemeye sistem tercihini al
    const stored = localStorage.getItem('theme');
    const prefersDark = window.matchMedia('(prefers-color-scheme: dark)').matches;
    this.isDark.set(stored === 'dark' || (!stored && prefersDark));

    // Theme değişimini DOM'a uygula
    effect(() => {
      document.documentElement.classList.toggle('dark-mode', this.isDark());
      localStorage.setItem('theme', this.isDark() ? 'dark' : 'light');
    });
  }

  toggle() {
    this.isDark.update(v => !v);
  }
}

// Component
@Component({...})
export class ThemeToggle {
  protected theme = inject(ThemeService);
}

// Template
<button matIconButton (click)="theme.toggle()">
  <mat-icon>{{ theme.isDark() ? 'light_mode' : 'dark_mode' }}</mat-icon>
</button>

```

□ Custom Palette

```
// styles.scss
@use '@angular/material' as mat;

// Kendi renk paleti tanımla (M3 tonal scale)
$tse-palette: (
  0: #000000,
  10: #001b3d,
  20: #003063,
  30: #00468b,
  40: #1e3a8a,    // ← ana renk (40 = base)
  50: #2563eb,
  60: #3b82f6,
  70: #60a5fa,
  80: #93c5fd,
  90: #dbeafe,
  95: #eff6ff,
  99: #fafbff,
  100: #ffffff,
);

html {
  @include mat.theme((
    color: (
      theme-type: light,
      primary: $tse-palette,
    ),
  ));
}
```

Angular CDK — Düşük Seviye Primitives

Material'ın iskeleti. Kendi component'ini yazarken kullan.

□ CDK Nedir?

Component Dev Kit: Material'ın altyapısı. Stil olmadan davranışlar. Angular Material kullanmasan bile CDK'yı kullanabilirsin. Çok güçlü.

□ Virtual Scrolling

```
import { ScrollingModule } from '@angular/cdk/scrolling';

@Component({
  imports: [ScrollingModule],
  template: `
    <cdk-virtual-scroll-viewport itemSize="50" class="viewport">
      @for (item of items(); track item.id) {
        <div class="item">{{ item.name }}</div>
      }
    </cdk-virtual-scroll-viewport>
  `,
  styles: [
    `.viewport { height: 400px; }
    .item { height: 50px; padding: 10px; }`
  ]
})
export class VirtualListComponent {
  // 100,000 item bile sorunsuz render!
  items = signal(generateItems(100_000));
}
```

□ Overlay

```

import { Overlay, OverlayRef } from '@angular/cdk/overlay';
import { TemplatePortal } from '@angular/cdk/portal';

@Component({...})
export class TooltipExample {
  private overlay = inject(Overlay);
  private viewContainerRef = inject(ViewContainerRef);

  @ViewChild('tooltipTemplate') tooltipTemplate!: TemplateRef<any>;

  showTooltip(origin: HTMLElement) {
    const overlayRef = this.overlay.create({
      positionStrategy: this.overlay.position()
        .flexibleConnectedTo(origin)
        .withPositions([
          {
            originX: 'center',
            originY: 'bottom',
            overlayX: 'center',
            overlayY: 'top',
            offsetY: 8,
          },
        ]),
      scrollStrategy: this.overlay.scrollStrategies.reposition(),
    });

    const portal = new TemplatePortal(this.tooltipTemplate, this.viewContainerRef);
    overlayRef.attach(portal);

    return overlayRef;
  }
}

```

□ Drag & Drop


```

import { DragDropModule, CdkDragDrop, moveItemInArray } from
 '@angular/cdk/drag-drop';

@Component({
  imports: [DragDropModule],
  template: `
    <div cdkDropList (cdkDropListDropped)="drop($event)">
      @for (task of tasks(); track task.id) {
        <div cdkDrag class="task">
          <span cdkDragHandle></span>
          {{ task.title }}
        </div>
      }
    </div>
  `
})
export class TaskListComponent {
  tasks = signal<Task[]>([...]);

  drop(event: CdkDragDrop<Task[]>) {
    this.tasks.update(arr => {
      const copy = [...arr];
      moveItemInArray(copy, event.previousIndex, event.currentIndex);
      return copy;
    });
  }
}

```

Diğer CDK Modülleri

Modül	Ne için
ally	Accessibility utilities (focus trap, live announcer)
bidi	RTL/LTR direction detection
clipboard	Copy to clipboard
layout	Breakpoint observer (responsive)
platform	Platform detection (iOS, Android, etc)
stepper	Stepper logic (no UI)
table	Table primitives
tree	Tree primitives

CDK'nın Gücü

PrimeNG, ng-bootstrap gibi üçüncü parti kütüphanelerin çoğu CDK üzerine kurulu. CDK'ı öğrenmek = framework'ün altını anlamak. Her zaman işine yarar.

☐ Modül 31 Özeti

Angular Material ile artık tam donanımdasın. Modern M3 design system, temel componentler, form'lar, navigation, theming ve CDK primitives — hepsi cebinde.

☐ Neler Öğrendin?

- ✓ Material Design 3 (M3) ve evrim
- ✓ Angular Material v20 kurulum
- ✓ Temel componentler (Button, Card, Icon, Chip)
- ✓ Form components (Input, Select, DatePicker)
- ✓ Overlay services (Dialog, SnackBar, BottomSheet)
- ✓ Navigation (Toolbar, Sidenav, Tabs, Stepper)
- ✓ M3 theming + dark mode + custom palette
- ✓ Angular CDK — virtual scroll, overlay, drag-drop

☐ Devam Et

Modül 32 — Tailwind CSS + Headless UI. Component library yerine utility-first yaklaşımla kendi UI'larını yazma.